



MAHINDRA UNIVERSITY HYDERABAD

École Centrale School of Engineering

Minor-I Examinations, September 2025

Program: B.Tech.

Branch: CM (2023 Batch)

Year: 3rd

Semester: I

Subject: Financial Mathematics (MA3116)

Date: 10/09/2025

Time Duration: 1.5 Hours

Start Time: 02:00 PM

Max. Marks: 30

Instructions:

1. Answer all questions.
2. Using scientific calculator is allowed.

Q 1:

Marks : 10

a) Define the following : (i) Asian Option, (ii) American Put Option, (iii) European Call Option, (iv) Forward contract.

Q 2:

Marks : 10

a) Consider a 3-period binomial model of asset pricing. Let the initial stock price be $S_0 = 10$ per share, $u = 2$ be up factor, $d = 0.5$ be down factor, $r = 0.25$ be rate of interest per time period, $K = 15$ be strike price.

(i) Compute the look-back option price process.

(ii) Let $\{S_n\}$ denote the stock price process and $\{M_n\}$ denote maximum till date stock price process. Then show that $\{(S_n, M_n)\}$ is a two dimensional Markov process.

Q 3:

Marks : 10

a) Consider a 3-period binomial model of pricing American option. Let the initial stock price be $S_0 = 8$ per share, $u = 2$ be up factor, $d = 0.5$ be down factor, $r = 0.25$ be rate of interest compounding per time period, $K = 8$ be strike price.

(a) Find the initial price of the American put option.

(b) Compute option delta process : $\Delta_0, \Delta_1, \Delta_2$.